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Education

Present 04.2026	Doctoral in Doctoral Student in Energy, École Polytechnique Fédérale de Lausanne Distributed Electrical Systems Laboratory
03.2026 09.2023	Master in Energy Science and Technology, École Polytechnique Fédérale de Lausanne Section of Electrical and Electronical Engineering GPA: 5.66/6.0
06.2023 09.2019	Bachelor in Energy Storage Science and Engineering, Xi'an Jiaotong University Faculty of Electronic and Information Engineering GPA: 91.16/100

Work Experience

Present 04.2026	Doctoral Assistant (100%) @ <u>Distributed Electrical Systems Laboratory</u> ➤ Contribute to laboratory research. ➤ Contribute to tasks related to the missions education, research, and innovation.
08.2025 02.2025	Internship in Power Systems of the Future (100%) @ <u>Hitachi Energy</u> ➤ Developed a Python library (MOSAIC) from scratch for large-scale capacity expansion analysis and optimization of future energy systems, formulated as Linear Programming (LP) problems and solved via Gurobi. The library features accelerated computation, integrated result processing, robust extendability, and clear design documentation, serving as a foundational tool of the team for energy system modeling. ➤ Enhanced post-processing and visualization workflows for a MATLAB-based capacity expansion model (ECOTRANS), applying the refined toolkit to multiple strategic studies. ➤ Provided continuous technical support on coding across the team.
09.2024 07.2024	Summer Internship in Charging System (100%) @ <u>SAIC Motor</u> ➤ Performed state-of-the-art review of on-board chargers, conducted a feasibility study on the vertical integration of on-board chargers. ➤ Designed control systems and simulation models for a 6.6 kW/400 V totem-pole bridge-less PFC converter and LLC resonant converter demo.
09.2022 03.2022	Internship in Energy Storage Market (40%) @ <u>China Energy Storage Alliance (CNESA)</u> ➤ Maintained and updated CNESA's energy storage project, policy and bidding information databases. ➤ Collected energy storage news articles; edited drafts of research reports and compiled quarterly global energy storage market tracking report.

Skills & Languages

Relevant Skills	Convex optimization(Gurobi, MOSEK, COPT, HIGHS), Power system analysis (pandapower, PyPSA), Smart grid technologies, Microgrid modeling and control, Energy system modeling
Other Skills	Model predictive control, Data analysis, Git for version control, $\text{\LaTeX}$, Life cycle assessment, MS Office
Coding	Python, MATLAB & Simulink, LabVIEW, HTML, CSS, JavaScript, Julia, Go
Language	Chinese — Native English — C1 Français — A2

Projects

- › Master thesis at DESL – A Two-Layer MPC Framework with Workload Flexibility and Energy Valorization for Real-Time Control of Sustainable Data Centers: Designing a two-layer model predictive control (MPC) framework to optimize real-time operations in sustainable data centers, focusing on maximizing energy valorization while minimizing the carbon footprint and operational costs. The energy consumption of the data center is modeled as MINLP problems and linearized for control design (MILP), which features an upper-layer for coordinating market bidding and a lower-layer for controlling flexible workloads, PV, BESS, and waste heat recovery systems.
- › Semester project at DESL – A Framework for Carbon Flow Tracing in Swiss Power Distribution Grids: Developed a framework for carbon footprint tracing from inferred Swiss low-voltage and medium-voltage grids. Modeled the generation (with the meteorology and roof suitability of PV panels) and load (with land-use categories) profiles. Developed customized average participation method with loss treatment for flow tracing. The framework is applied using both a small toy example and a synthetic distribution grid in CH.
- › Semester project at DESL – Lithium-Ion Cell Equivalent Circuit Model Parameters Estimation from Time-Domain Measurements: Utilized voltage and current signals sampled by battery management system for estimating the parameters of battery equivalent circuit models. Explored the identifiability of different parameters and developed a low-frequency model for estimating constant phase element via grey-box estimation.
- › My Personal Website[🔗]: Built my personal website wenyuliu.ch with Quarto, deployed by GitHub Pages. Hosted my CV, journey and some projects on the website for public reference.
- › The Impact of COVID-19 on Green Energy Transition in European Countries[🔗]: Delved into the interplay between the COVID-19 and renewable transition in 6 EU countries, analyzed datasets on generation, interests, mobility and government intervention, uncovered how the pandemic impacted electricity generation, consumption, and awareness of renewable energy.

Publications

- › **W. Liu**, E. Figini, and M. Paolone, “Carbon flow tracing in distribution grids: from method to application,” *Electric Power Systems Research*, vol. 262, p. 113548, Jan. 2027, doi: 10.1016/j.epsr.2026.113548. Available: <https://www.sciencedirect.com/science/article/pii/S0378779626008412>
- › *Submitted and under review*: **W. Liu**, E. Figini, and M. Paolone, “Real-Time Control of Sustainable Data Centers: A Two-Layer Model Predictive Control Framework with Workload Flexibility and Heat Recovery,” 2026, doi: 10.48550/arXiv.2608.16432

Honors & Awards

- › **Prix de l'État de Vaud (2026)**, awarded for a Master's project presenting an original solution promoting the efficient use of energy.
- › **Graduate with Honors (2023)**, Qian Xuesen College, Xi'an Jiaotong University.
- › **First-Class Academic Scholarship (2020)**, Xi'an Jiaotong University.
- › **Outstanding Student Award (2020)**, Xi'an Jiaotong University.